SOFTWARE ARCHITECTURE & PROJECT SPECIFICATION

COMPREHENSIVE PROJECT REPORT & BLUEPRINT

PG Booking & Management Platform

A centralized, multi-tier digital ecosystem uniting PG owners, administrators, and guests through real-time bed inventory, automated WhatsApp communications, cross-platform mobile & web applications, and optional instant UPI settlement.

- Admin Enterprise Panel
- iOS & Android Mobile Apps
- Customer Responsive Web App
- WhatsApp Business API Integration
- Optional UPI Payment Gateway
- Role-Based Access Control

DOCUMENT VERSION	TARGET DEPLOYMENT	PREPARED FOR	CLASSIFICATION
v1.0 (Production Final)	Enterprise Cloud (AWS)	Client Evaluation & Sign-off	Confidential / RFP Spec

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25 STRUCTURED SECTIONS

This document constitutes the definitive software project proposal, system requirements specification (SRS), technical architecture blueprint, and commercial estimate for the **PG Booking & Management Platform**.

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Document Revision & Control History

Version	Release Date	Author / Role	Summary of Changes / Scope
v0.1 Draft	Oct 12, 2026	Lead Business Analyst	Initial stakeholder requirement gathering & feature catalog.
v0.5 Architecture	Oct 20, 2026	Principal Solutions Architect	Technical stack formulation, database ER design, WhatsApp API specs.
v1.0 Final	Oct 30, 2026	Technical Delivery Director	Client-ready commercial proposal, milestone schedule, modular UPI scope.

📌 **Executive Reading Guide**

This report is partitioned into technical, operational, and commercial sections. Stakeholders focusing on financial feasibility may review **Section 18 & 19** (Timeline & Cost), while engineering teams should consult **Section 14 & 15** (Architecture & Database).

01 Executive Summary

STRATEGIC OVERVIEW

The **PG Booking & Management Platform** is a next-generation software ecosystem created to organize and automate the residential Paying Guest (PG) and co-living rental sector. The accommodation market in urban education and tech centers faces immense friction caused by unorganized paper registers, delayed lead responses, cash rent disputes, and opaque bed vacancies. This platform bridges property owners, administrative wardens, and tenants through a synchronized digital platform.

🖥️ Admin Management Panel

WEB SPA

Comprehensive operational hub for multi-property oversight, real-time bed-level inventory tracking, tenant KYC verification, booking approvals, and automated reporting.

📱 Customer Mobile App

IOS / ANDROID

High-conversion cross-platform mobile app featuring GPS proximity search, 360 photo galleries, bed-level selection, WhatsApp chat, and booking tracking.

🌐 Customer Web Portal

RESPONSIVE SSR

SEO-optimized discovery portal offering search engine indexing for localized PG queries, instant booking requests, and full account self-service.

Core Strategic Differentiators

1. Conversational WhatsApp-First Strategy

Harnesses India's most ubiquitous messaging channel for instant click-to-chat inquiries alongside automated transactional messaging via the official WhatsApp Business Cloud API (booking alerts, approvals, rent reminders, and check-in guides).

2. Modular & Decoupled UPI Payment Layer

Architecture supports zero-friction onboarding where bookings function as offline payment requests, while retaining an optional plug-and-play UPI payment gateway (QR codes, deep-links, automated reconciliation) toggled via a single feature flag.

Key Expected Business Outcomes & KPIs

35% ↓

Bed Vacancy Duration

Faster inventory turnaround via real-time bed visibility.

< 30s

Lead Response Time

Instant automated WhatsApp greetings and manager routing.

60% ↓

Administrative Overhead

Automated monthly rent invoicing and digital KYC vaults.

100%

Booking Accuracy

Zero double-bookings through atomic bed-level locking.

💼 Commercial & Timeline Summary

The core platform is estimated at **1,000 engineering hours** over a **12 to 14 week timeline** with a core software investment of **₹7,50,000 INR**. The optional UPI integration is priced separately at **₹65,000 INR**, enabling flexible budget allocation.

02 Project Introduction

CONTEXT & BACKGROUND

The massive growth of corporate tech corridors, medical universities, and higher education hubs across tier-1 and tier-2 cities has driven unprecedented demand for flexible, safe, and cost-effective student and working-professional housing. Traditionally, PG properties have operated as unorganized, standalone units managed through physical paper registers, unverified phone calls, and manual cash handling.

The **PG Booking & Management Platform** transitions this sector into a synchronized digital environment. By standardizing property onboarding, visual asset cataloging, bed availability tracking, booking request workflows, and tenant record management, the platform eliminates operational friction for property owners while delivering a transparent, frictionless experience for guests.

03 Problem Statement

MARKET INEFFICIENCIES

The existing operational framework in the PG ecosystem suffers from systemic inefficiencies impacting both property operators and prospective tenants:

Operational Domain	Traditional Manual Process	Operational & Financial Impact
Inventory & Bed Tracking	Physical register notebooks or rudimentary spreadsheets; bed vacancy updated verbally between staff.	Frequent double-booking errors, delayed vacancy updates, lost revenue from unsold beds, zero centralized visibility.
Tenant Lead Conversion	Unanswered phone calls, lost WhatsApp messages on personal phones, delayed photo sharing.	Over 40% lead drop-off; prospective guests book competitors who respond faster with verified photos and pricing.
KYC & Document Storage	Physical photocopies of Aadhaar/ID cards stored in filing cabinets; no expiry or verification checks.	Severe legal non-compliance risks, vulnerability to fire/theft, inability to rapidly respond to local police verification mandates.
Rent & Deposit Collection	Cash payments or unlinked bank transfers; manual receipt books; manual payment reconciliation.	Frequent disputes over security deposit deductions, untracked overdue rents, manual accounting errors, delayed cashflow.
Tenant Experience	Mandatory physical visits just to check room layout; lack of written house rules or transparent refund terms.	High frustration among working professionals; disputes regarding notice periods and electricity/utility bills.

> ⚠️ **The Core Operational Bottleneck**
> Without centralized digital infrastructure, property owners cannot scale past 1 or 2 buildings without experiencing catastrophic operational leakage, uncollected rental dues, and regulatory non-compliance.

04 Project Objectives & 05 Proposed Solution

SMART ARCHITECTURE

04. Core Project Objectives

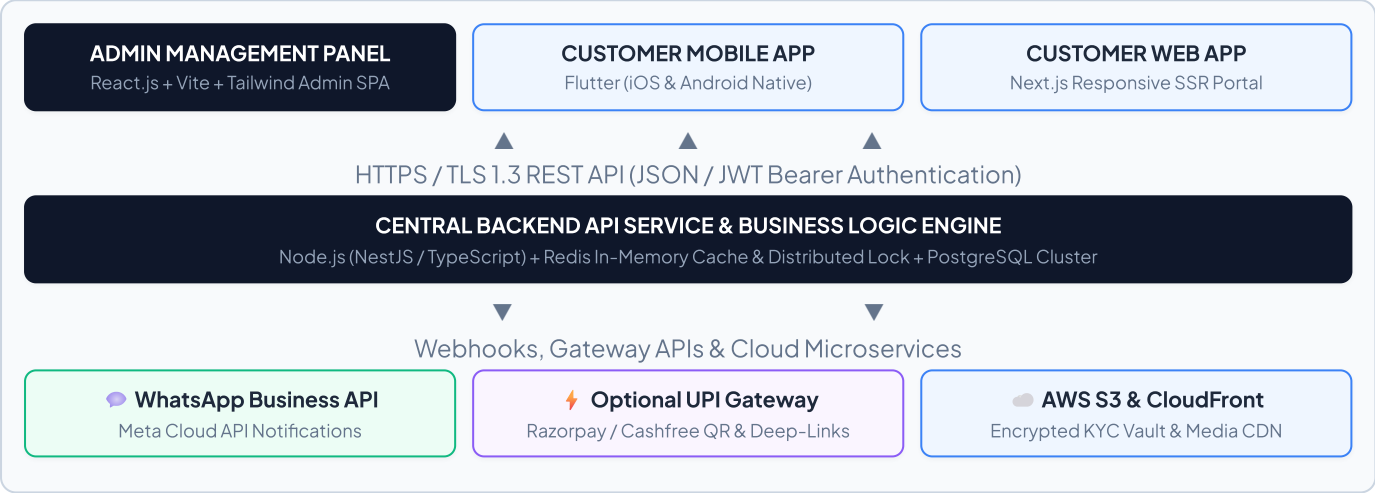
1. Real-Time Bed Inventory
100% synchronization of room and individual bed availability across multiple properties, eliminating overbooking.

2. WhatsApp Conversions
Accelerate lead response times to under 30 seconds via automated WhatsApp greeting templates and direct chat links.

3. Frictionless Booking
Enable customers to complete a digital booking request in under 2 minutes through mobile and web channels.

05. Proposed Solution: Multi-Tier Ecosystem Architecture

The platform connects three client endpoints to a central microservices-ready cloud API, orchestrating conversational messaging, secure storage, and optional financial gateways:



Architectural Principle: Atomic Bed Locking

When a customer submits a booking request, Redis places an atomic reservation lock on the specific bed identifier for a configurable window (e.g., 2 hours). This strictly prevents concurrent race conditions where two users attempt to reserve the same bed.

06 Project Scope & 07 System Modules

FUNCTIONAL BASELINE

06. Scope Classification Matrix (Core vs Optional)

Project deliverables are cleanly categorized into Core Baseline and Optional Add-ons to maintain budget transparency:

Scope Category	Modules & Capabilities Included	Classification
Admin Web Dashboard	Property catalog, room/bed inventory, booking approval inbox, tenant KYC vault, cash rent ledger, reports, RBAC.	CORE SCOPE
Customer Mobile App	Flutter iOS & Android apps, phone OTP auth, GPS search, filter engine, property gallery, bed selection, WhatsApp chat.	CORE SCOPE
Customer Web App	Next.js responsive web app, SEO localized landing pages, bed availability viewer, booking requests, tenant account.	CORE SCOPE
WhatsApp Messaging	Direct wa.me click-to-chat links + Meta WhatsApp Business Cloud API automated transactional triggers (alerts, confirmations).	CORE SCOPE
UPI Payment Integration	Dynamic UPI QR generation, mobile UPI deep-linking (GPay/PhonePe), payment webhook verification, automated receipt generation.	OPTIONAL ADD-ON

07. System Modules and Functional Matrix

1. Auth & Access Control
Phone OTP login, JWT token lifecycle, password encryption (Argon2), and role-based guards.

2. Property Catalog
Property master, geo-coordinates, branch tagging, amenity checklist, gallery upload, house rules.

3. Room & Bed Engine
Hierarchical (Property -> Floor -> Room -> Bed). Manages room types, pricing, and live bed states.

4. Booking Engine
Inbound request -> Inventory hold -> Admin approval/rejection -> Check-in schedule -> Room assignment.

5. Tenant Record Vault
Tenant profiles, encrypted government KYC upload (Aadhaar, Passport), emergency contacts, stay logs.

6. Payment & Invoicing
Automated monthly rent cron, deposit ledger, manual cash tracking, optional UPI, PDF receipts.

7. WhatsApp Hub
Webhook listening, template message dispatches, transactional status notifications, wa.me routing.

8. Analytics & Reporting
Occupancy trends, revenue aggregations, outstanding rent aging lists, 1-click PDF/Excel export.

9. System Settings
Platform global variables, API key managers (WhatsApp/UPI), notification copy, administrative audit trails.

08 Admin Management Panel (Part 1: Properties & Rooms)

CORE PLATFORM

The Admin Management Panel serves as the central command dashboard for PG owners and operational staff. Built as a high-performance Single Page Application (SPA), it provides instantaneous access to operational and financial data.

8.1 Real-Time Operations Dashboard

🏢 Properties & Capacity
Total properties under management, aggregate bed capacity, total occupied beds, and overall occupancy %.

📥 Booking Funnel
New booking requests received today, pending review requests, confirmed bookings, and cancellations.

💵 Financial Snapshot
Rent collected current month, outstanding rent dues, pending security deposits, and cash receipts to verify.

⚡ Live Activity Feed
Chronological audit log: "Guest Rahul Sharma booked Room 204", "Bed 3B marked checked-in by Warden Amit".

8.2 PG Property Management

Enables full lifecycle configuration of PG buildings across multiple branches:

- **Property Profile & Location:** Name, description, postal address, landmark, and GPS latitude/longitude integrated with Google Maps.
- **Categorization & Rules:** Classification (Men, Women, Co-living/Unisex). Amenity tags (Wi-Fi, Food, Power Backup, RO Water, Biometric Entry, Washing Machine, Housekeeping). Written curfew times, visitor policies, and notice period requirements.
- **Photo Gallery CDN:** Drag-and-drop multi-image uploader with WebP compression and CDN distribution.

8.3 Room & Bed Inventory Engine

Provides granular, bed-level inventory tracking to eliminate duplicate bookings:

- **Hierarchical Organization:** Property ➔ Wing/Block ➔ Floor ➔ Room Number ➔ Bed Identifier.
- **Room Configurations:** Single Private, Double Sharing, Triple Sharing, Four Sharing, and Dormitory formats.
- **Granular Bed Identifiers:** Unique alphanumeric tags per bed (e.g., `Room 201-Bed A`, `Room 201-Bed B`).
- **Financial Attributes per Bed:** Base monthly rent, maintenance fees, utility surcharges, and refundable security deposit.

Bed Operational Status Lifecycle

Bed Status	Visual Indicator	System Behavior & Booking Availability
Available	GREEN BADGE	Visible to public on Mobile and Web apps; open for instant booking requests.
Reserved / Locked	AMBER BADGE	Booking request under review; temporarily blocked in Redis to prevent concurrent bookings.
Occupied	BLUE BADGE	Active resident assigned; linked to current tenant profile and monthly billing schedule.
Maintenance	RED BADGE	Bed undergoing repairs/painting; excluded from public availability until cleared by admin.

08 Admin Panel (Part 2: Tenants, Invoicing & Reports)

CORE PLATFORM

8.4 Booking Lifecycle Management Workflow

Stage	Admin Action & Capability	System Automation Trigger
Request Intake	Review incoming guest request: name, phone, chosen PG, room type, desired bed, move-in date.	Automated WhatsApp acknowledgment to guest; Admin alert dispatched.
Evaluation	Admin evaluates bed availability; clicks **Approve** or **Reject** (with note). Can reassign bed.	Bed switches to `Reserved`; Instant WhatsApp confirmation template sent.
Check-In	Warden verifies tenant KYC documents, records advance deposit, and marks **Checked-In**.	Bed switches to `Occupied`; Active Tenant profile activated in system.
Check-Out	Admin initiates vacate process, records room inspection, calculates deductions, marks **Checked-Out**.	Bed immediately reverts to `Available`; Tenant profile moved to archive.

8.5 Tenant Management & KYC Document Vault

Provides centralized digital record management for all active and past residents:

- **Comprehensive Resident Profiles:** Full name, permanent address, contact numbers, blood group, university/employer details, and mandatory emergency contact names/numbers.
- **Encrypted Digital KYC Vault:** Upload and store government ID proofs (Aadhaar Card, PAN Card, Passport, Driving License) and student/work ID cards in private S3 buckets with expiring pre-signed URLs.
- **Tenancy Filtering:** One-click views for `Active Residents`, `Notice Period Active`, and `Past / Alumni Tenants`.

8.6 Invoicing, Rent & Payment Management

Automated Monthly Rent
Scheduled monthly cron generates rent invoices for all active tenants on the 1st of every month with pre-set due dates.

Dual Payment Tracking
Tracks both online UPI transactions and manual cash / NEFT bank entries with transaction reference recording.

Security Deposit Ledger
Tracks initial deposit received, itemized damage/painting deductions, and final refund disbursement status.

8.7 WhatsApp Communication Hub, Reports & Settings

Admins can view message delivery logs (Sent, Delivered, Read), launch direct 1-to-1 WhatsApp chats, export financial/occupancy reports to Excel/PDF, and manage staff roles (Wardens, Accountants) with granular RBAC permissions.

> 📄 **One-Click Export Suite**
> All tables (Occupancy Trends, Monthly Collections, Overdue Rent Aging, Tenant Registers) support instant 1-click export into formatted Excel spreadsheets (.xlsx) and printable PDF documents for audit and accounting compliance.

09 Customer Mobile Application (Android & iOS)

CORE PLATFORM

The Customer Mobile Application is architected using **Flutter** to deliver a fluid, native 60fps experience across both Android and iOS devices from a unified codebase. The app prioritizes intuitive discovery, transparent room previews, instant WhatsApp communication, and hassle-free booking.

🔐 Frictionless Authentication

Mobile number login with instant 6-digit SMS OTP verification. No complex passwords required. Returning users remain securely logged in using persistent refresh tokens.

📍 Smart Geo-Discovery & Search

Auto-detects user GPS location to show nearby PGs. Search bar with autocomplete for tech parks, universities, metro stations, and popular neighborhood landmarks.

🔍 Comprehensive Filter Engine

Filter by Gender (Men / Women / Co-living), Monthly Budget Slider, Room Type (Single, 2-sharing, 3-sharing), AC / Non-AC, Food Included, and Specific Amenities.

📸 Immersive Property Showcase

High-resolution photo galleries with full-screen pinch-to-zoom, verified amenity icon grids, house rules, curfew timings, and integrated Google Maps route directions.

Interactive Bed Availability Visualizer

Unlike generic rental apps that display vague room counts, this application shows **exact bed-level availability**. Users can tap on a room type (e.g., "Deluxe 2-Sharing Room 204") to see a visual layout showing Bed A (Occupied) and Bed B (Available). The user selects Bed B, picks their planned move-in date, and submits a booking request in two taps.

Booking Management & Live Status Tracker

Screen / Feature	Functionality Description	User Benefit
Active Booking Card	Displays current booking status with a visual timeline tracker: `Request Sent` ➔ `Under Review` ➔ `Approved` ➔ `Confirmed`.	Complete clarity on booking progress without needing follow-up phone calls.
Check-In Pass	Once approved, generates a digital check-in summary card containing property address, warden phone number, bed number, and check-in instructions.	Frictionless arrival experience; quick contact with property staff on site.
Booking Cancellation	Allows customers to cancel pending requests directly in-app with an optional reason note.	Transparent cancellation with immediate freeing of reserved inventory.
Resident Portal Mode	Once checked in, the app switches to Resident Mode: view monthly rent invoices, download payment receipts, and submit notice of vacating.	Single unified app serving both prospective leads and active long-term residents.

💬 Floating WhatsApp Quick-Action

A persistent "Chat with PG Manager on WhatsApp" floating button is visible on every property screen. Tapping launches WhatsApp with prefilled room details, enabling instant 1-to-1 conversation with zero friction.

10 Web Application & 11 WhatsApp Integration

CORE PLATFORM

10. Customer Responsive Web Application

The Customer Web Application is engineered using **Next.js**, combining lightning-fast Server-Side Rendering (SSR) for Search Engine Optimization (SEO) with responsive client-side interactivity across desktop, tablet, and mobile browsers:

🌐 SEO Landing Pages

Programmatic landing pages optimized for search engines (e.g., `/pg-in-koramangala`, `/pg-in-hitec-city`) to drive organic customer acquisition.

💻 Split-Screen Desktop UI

Rich desktop layout featuring interactive map on the left with property cards on the right, mirroring international travel portals.

⚡ Zero App Barrier

Users discovering properties via Google Search can inspect rooms, check bed vacancies, and submit booking requests instantly without app install.

11. WhatsApp Messaging Integration (Two-Tier Architecture)

Tier 1: Direct WhatsApp Click-to-Chat

Universal deep links (`https://wa.me/91XXXXXXXXXX?text=...`). Launches WhatsApp on the user's device with a pre-filled enquiry: *"Hello! I am interested in Room 204 at Green View Luxury PG."*

Tier 2: WhatsApp Business Cloud API

Programmatic event-driven transactional notifications triggered by backend webhooks using pre-approved Meta message templates for booking alerts, confirmations, and rent reminders.

Approved Transactional WhatsApp Message Templates

Template Event	Trigger Condition	Message Content Delivered to Customer
Booking Request Received	Customer submits booking request on Mobile or Web.	*"Hi {{1}}, your booking request for {{2}} at {{3}} has been received! Our team is reviewing bed availability. Ref: #{{4}}."*
Booking Confirmed	Admin clicks "Approve" in Admin Panel.	*"Congratulations {{1}}! Your booking for Bed {{2}}, Room {{3}} at {{4}} is APPROVED. Check-in Date: {{5}}. Warden Contact: {{6}}."*
No Vacancy Alert	Admin rejects request due to full occupancy.	*"Hello {{1}}, unfortunately the requested bed at {{2}} is full. We have recommended alternate nearby rooms here: {{3}}."*
Check-In Reminder	24 hours before scheduled move-in.	*"Hi {{1}}, we look forward to welcoming you to {{2}} tomorrow! Directions: {{3}}. Please carry original Aadhaar card."*
Monthly Rent Due	Triggered on 1st & 5th of month for active tenants.	*"Dear {{1}}, your monthly rent of ₹{{2}} for Room {{3}} is due on {{4}}. Pay at reception or via your app: {{5}}. Thank you!"*

12 Optional UPI & 13 User Roles Matrix

MODULAR ARCHITECTURE

12. Optional UPI Payment Integration Architecture

The UPI payment infrastructure is designed as an **isolated, modular micro-module** controlled by a platform feature flag (`ENABLE_UPI_PAYMENT = true/false`). The core booking lifecycle functions seamlessly whether this module is enabled or disabled:

Dynamic UPI QR Codes
Generates unique dynamic QR code encoded with exact invoice amount and merchant VPA. Scannable by any UPI app (GPay, PhonePe, Paytm, BHIM).

UPI App Deep-Linking
On Mobile App, clicking "Pay via UPI" opens native app chooser (Google Pay, PhonePe). Seamlessly executes payment and returns.

Webhook Reconciliation
Gateway sends HMAC-SHA256 signed webhook to backend server. Server validates signature, marks invoice PAID, and generates receipt PDF.

Dual-Mode Flexibility

When UPI is Disabled: The platform operates in "Pay at Property / Offline Reconciliation" mode. Admins manually mark rents as paid upon receiving cash/NEFT.

When UPI is Enabled: Supports booking token advances (e.g., ₹1,000), security deposit collections, and in-app 1-tap monthly rent payments.

13. Role-Based Access Control (RBAC) Permissions Matrix

System Functionality	Super Admin	Property Admin	Warden / Staff	Customer / Tenant
Add/Edit/Delete Properties	Full Access	Assigned Only	No Access	No Access
Configure Room Pricing & Deposits	Full Access	Assigned Only	No Access	No Access
Approve / Reject Bookings	Full Access	Assigned Only	Assigned Only	No Access
Check-In & Check-Out Guests	Full Access	Assigned Only	Assigned Only	No Access
View Financial Reports & Revenue	Full Access	Assigned Only	No Access	No Access
Manage Staff Accounts & Roles	Full Access	No Access	No Access	No Access
Submit Booking Requests & Enquiries	N/A	N/A	N/A	Full Access
View Own Invoices & Download Receipts	N/A	N/A	N/A	Own Records Only

14 Technical Architecture & Technology Stack

ENGINEERING BLUEPRINT

The platform is engineered following a modern, cloud-native, microservices-ready monolithic architecture designed for rapid development velocity, horizontal scalability, and low operational maintenance overhead.

Layer / Component	Recommended Technology	Architectural Rationale & Advantages
Mobile Application	**Flutter (Dart)**	Single codebase for native iOS and Android; compiles to ARM machine code; delivers fluid 60fps UI animations; eliminates dual iOS/Android development costs.
Customer Web App	**Next.js (React / SSR)**	Server-Side Rendering (SSR) ensures full SEO indexing for localized PG searches; lightning-fast initial page load; automatic image optimization.
Admin Dashboard	**React.js + Vite + TailwindCSS**	Rapid, responsive SPA dashboard; component-based architecture; rich data-table libraries and real-time state synchronization via React Query.
Backend API Server	**Node.js (NestJS / TypeScript)**	Enterprise-structured framework; native TypeScript type safety; asynchronous non-blocking event loop ideal for high-concurrency booking requests and webhooks.
Primary Database	**PostgreSQL (AWS RDS)**	Rock-solid relational integrity for financial ledgers, inventory bed tracking, and transactional ACID compliance; JSONB support for dynamic property amenities.
In-Memory Cache & Lock	**Redis**	Session storage, OTP rate-limiting, and distributed locking to guarantee zero concurrent double-booking of the same bed identifier.
Cloud Infrastructure	**Amazon Web Services (AWS)**	Docker containers on AWS ECS/Fargate or EC2; S3 for asset/KYC storage; CloudFront CDN for global low-latency media delivery; Route 53 DNS.
Messaging Gateway	**Meta WhatsApp Cloud API**	Direct enterprise integration with Meta servers; high throughput; official green-badge template support; low per-conversation messaging costs.
Optional Payment Gateway	**Razorpay / Cashfree API**	Leading Indian payment processors; robust UPI intent, QR codes, automated webhook signatures, and direct bank settlement.

Scalability Target

Supports 100,000+ monthly active users and 10,000+ concurrent bed listings with <150ms average API response times.

Containerized DevOps

Docker container images with multi-stage builds deployed via GitHub Actions CI/CD to AWS ECS with zero downtime.

Disaster Recovery

Automated daily PostgreSQL snapshots on AWS RDS with 30-day retention and multi-AZ failover capability.

15 Database Schema & REST API Overview

DATA ARCHITECTURE

Relational Entity Schema Definitions (PostgreSQL)

`properties`
id, name, slug, gender_type, address, city, lat, lng, amenities, rules, status, created_at.

`rooms`
id, property_id, room_no, floor, room_type, base_rent, deposit_amount, status.

`beds`
id, room_id, bed_label (e.g. 'A'), is_occupied, current_tenant_id, status.

`bookings`
id, booking_ref, user_id, bed_id, move_in_date, status, token_amount, payment_mode.

`tenants`
id, user_id, bed_id, checkin_date, checkout_date, kyc_type, kyc_doc_url, emergency_phone.

`payments`
id, invoice_ref, tenant_id, amount, payment_type, upi_txn_id, status, receipt_pdf_url.

Core RESTful API Endpoint Catalog

HTTP	Endpoint Route	Description & Purpose	Auth Scope
`POST`	`/api/v1/auth/otp-request`	Dispatches 6-digit SMS OTP to user mobile.	PUBLIC
`POST`	`/api/v1/auth/otp-verify`	Validates OTP; returns JWT access & refresh tokens.	PUBLIC
`GET`	`/api/v1/properties`	List PGs with search, geo-distance, gender, budget filters.	PUBLIC
`GET`	`/api/v1/properties/:id/rooms`	Fetch rooms and individual bed availability for a PG.	PUBLIC
`POST`	`/api/v1/bookings`	Submit booking request; places temporary lock on bed.	CUSTOMER
`PATCH`	`/api/v1/admin/bookings/:id/status`	Approve or reject booking; triggers WhatsApp template.	ADMIN
`POST`	`/api/v1/payments/upi-order`	Generates dynamic UPI QR or intent order (Optional UPI).	OPTIONAL UPI
`POST`	`/api/v1/webhooks/whatsapp`	Receives inbound WhatsApp message status & chats.	SYSTEM WEBHOOK

OpenAPI 3.0 & Postman Deliverables

Every endpoint is fully documented via interactive Swagger/OpenAPI specifications. A ready-to-run Postman collection with environment variables will be handed over for seamless testing and client verification.

16 Security, Data Protection & 17 QA Methodology

COMPLIANCE & QUALITY

16. Security & Data Protection Controls

🔒 Transport & Encryption Standards

All data in transit is encrypted using TLS 1.3 with HSTS. Database volumes and S3 storage are encrypted at rest using AES-256. Sensitive customer KYC documents are stored in private cloud buckets with time-limited pre-signed URLs (15-minute expiry).

🛡️ Authentication & Session Lifecycle

Stateless JSON Web Tokens (JWT) with brief 15-minute validity paired with secure, rotating refresh tokens stored in HTTP-only, SameSite Secure cookies to eliminate Cross-Site Scripting (XSS) credential theft.

🛑 Attack Mitigation & Rate Limiting

Redis-backed token bucket rate limiters prevent SMS OTP bombing and brute-force attacks. Strict CORS whitelisting, parameterized SQL queries via TypeORM (preventing SQL injection), and Helmet.js HTTP header hardening.

⚖️ DPDP Act & Privacy Compliance

Compliant with India's Digital Personal Data Protection (DPDP) Act. Explicit consent collected before processing tenant phone numbers and Aadhaar documents. Right-to-be-forgotten mechanisms for archived tenants past statutory audit timeframes.

17. Development Methodology & QA Framework

The project executes under an **Agile Scrum Methodology** divided into 2-week sprint cycles, featuring sprint planning, daily standups, and bi-weekly client demo sessions:

Testing Discipline	Scope & Tooling	Quality Acceptance Standard
Unit & Integration Tests	Jest & Supertest validating business logic, rent calculations, and API endpoint responses.	Minimum 80% code coverage on core billing and inventory modules.
End-to-End (E2E) Testing	Playwright automated UI test scripts testing full booking flows on Web and Admin portals.	100% pass rate on booking submission, approval, and invoicing paths.
Mobile Device Testing	Testing across 15+ physical Android and iOS devices covering various screen ratios and OS versions.	Zero layout breaking; smooth 60fps animations on budget Android phones.
Security Vulnerability Scan	OWASP Top-10 audit, automated dependency scanning via Snyk, and manual penetration testing.	Zero Critical or High severity vulnerabilities prior to production launch.

18 Project Timeline and Milestones

14-WEEK ROADMAP

The total estimated development timeline for the **Core Platform** is **12 Weeks**, with an additional **2 Weeks** dedicated to the Optional UPI Payment Integration, User Acceptance Testing (UAT), and App Store submission, totaling **14 Weeks** from contract signing to production rollout:

Phase	Development Phase Name	Duration	Key Deliverables & Milestones
Phase 1	Requirement Analysis & System Specs	Weeks 1 – 2	Finalized SRS document, user personas, technical architecture blueprint, and API contracts.
Phase 2	UI/UX Design System & Wireframes	Weeks 2 – 4	Figma interactive clickable prototypes for Admin Dashboard, Customer Mobile App, and Web Portal.
Phase 3	Database Design & Backend Core APIs	Weeks 4 – 6	PostgreSQL schema deployment, Redis cache setup, JWT auth, property, room, and bed management APIs.
Phase 4	Admin Management Panel Development	Weeks 6 – 8	Operational dashboard, property/bed manager, booking approval inbox, tenant KYC vault, and offline ledger.
Phase 5	Customer Mobile App (Flutter)	Weeks 7 – 10	Cross-platform iOS and Android builds, phone OTP auth, GPS search, filter engine, bed selection, and booking tracker.
Phase 6	Customer Web Portal (Next.js)	Weeks 8 – 11	Responsive web interface, SEO landing pages, room availability checker, booking request form, and user hub.
Phase 7	WhatsApp Integration (Click & API)	Weeks 10 – 11	Meta Cloud API webhook configuration, template registration, automated transactional triggers, and wa.me links.
Phase 8	Optional UPI Payment Integration	Weeks 11 – 12	Razorpay/Cashfree gateway integration, dynamic QR generation, mobile UPI deep-linking, and receipt dispatches.
Phase 9	Quality Assurance, Security & UAT	Weeks 12 – 13	End-to-end regression testing, cross-browser/device validation, OWASP security audit, performance tuning.
Phase 10	Production Deployment & Handover	Week 14	AWS cloud setup, App Store & Google Play submission, admin staff training, documentation & code handover.

Key Milestone Sign-Off Schedule

🎯 **Milestone 1 (Week 4)**

Design Approval: Interactive Figma prototype sign-off for Mobile, Web, and Admin applications.

🎯 **Milestone 2 (Week 8)**

Admin & API Staging: Live backend APIs operational with functional Admin Panel managing properties and beds.

🎯 **Milestone 3 (Week 14)**

Production Launch: Mobile apps live on App Stores, Web live on domain, admin handover completed.

19 Project Cost Estimation

COMMERCIAL PROPOSAL

The commercial proposal provides transparent, modular pricing separating the **Core Platform Development** from the **Optional UPI Payment Module** and third-party recurring operating expenditures:

19.1 Core Platform Software Development Cost

Development Module / Component	Scope Summary	Estimated Hours	Investment (INR)
Admin Management Web Dashboard	SPA portal, property/bed manager, booking inbox, tenant KYC vault, reports.	180 hrs	₹1,35,000
Customer Mobile Application	Flutter iOS & Android apps, OTP login, GPS search, bed booking, WhatsApp.	220 hrs	₹1,65,000
Customer Web Application	Next.js responsive web app, SEO landing pages, bed availability viewer.	140 hrs	₹1,05,000
Backend REST APIs & Database	Node.js/NestJS, PostgreSQL schema, Redis caching, cron jobs, auth.	190 hrs	₹1,42,500
WhatsApp Messaging Integration	Direct wa.me deep links + Meta WhatsApp Business Cloud API webhooks.	70 hrs	₹52,500
UI/UX Design & Prototyping	Complete Figma design system, clickable prototypes for all 3 apps.	90 hrs	₹67,500
QA Testing, Security & Deployment	Automated testing, security audit, AWS cloud setup, App Store launch.	110 hrs	₹82,500
TOTAL CORE PLATFORM DEVELOPMENT		**1,000 hrs**	**₹7,50,000**

19.2 Optional Integration Modules (Priced Separately)

Optional Module Name	Functional Scope	Add-on Cost (INR)
UPI Payment Integration	Razorpay/Cashfree gateway integration, dynamic QR code generation, mobile UPI deep-linking, webhook signature validation, automated PDF receipt generator.	**₹65,000**
SMS Gateway Integration (Fallback)	Dual SMS integration (e.g. Fast2SMS/Twilio) as backup for booking alerts when WhatsApp is unreachable.	**₹20,000**

19.3 Estimated Recurring Third-Party Operating Expenses (Client Direct)

Service Component	Provider & Description	Estimated Recurring Cost
Cloud Infrastructure Hosting	AWS (t4g.small server, RDS PostgreSQL, S3, CloudFront)	Approx. ₹3,500 – ₹5,000 / month
WhatsApp Business Cloud API	Meta (First 1,000 service conversations free/month; Utility: ~₹0.35/msg)	Usage-based (Approx. ₹1,000 / month)
Developer Accounts	Google Play Console ($25 one-time) & Apple Developer ($99/year)	Approx. ₹10,500 first year

20 Deliverables & 21 Assumptions

HANDOVER INVENTORY

20. Project Deliverables Inventory

Upon final project acceptance and milestone completion, the client will receive complete ownership of all software assets, codebases, and documentation:

📦 Source Code Repositories

- Full Git repository for Admin Web Panel (React + Tailwind).
- Full Git repository for Customer Mobile App (Flutter).
- Full Git repository for Customer Web App (Next.js).
- Full Git repository for Backend REST API (Node.js/NestJS).

📱 Production Application Builds

- Production Android App Bundle (.aab) & signed APK.
- Compiled iOS build submitted to Apple TestFlight / App Store.
- Live deployed Admin Portal on client-designated cloud domain.
- Live deployed Customer Web Portal with SSL certificate.

📚 Technical Documentation

- Swagger / OpenAPI 3.0 interactive API documentation.
- Postman API collection with pre-configured environment tokens.
- Database Entity-Relationship (ER) diagram and migration scripts.
- Cloud deployment and Docker containerization runbooks.

🎓 User Training & Support Assets

- Comprehensive Admin Operations PDF User Manual.
- Property Warden quick-reference onboarding guide.
- 2 interactive virtual training sessions for administrative staff.
- 30-day post-launch warranty and bug-fixing support.

21. Assumptions and Dependencies

Key Project Assumptions

- **Client Review Cadence:** Client will provide feedback and milestone sign-offs within 3 business days of sprint reviews to prevent delivery delays.
- **Property Photography & Content:** High-resolution photos, property descriptions, and room inventory numbers will be supplied by the client during Phase 2.
- **Legal Policies:** House rules, terms of stay, notice periods, and privacy policies will be formulated and approved by client legal counsel.

Third-Party Dependencies

- **Meta WhatsApp Business Approval:** WhatsApp Business API requires a verified Meta Business Manager account. Approval timelines depend on Meta compliance checks (typically 2–4 business days).
- **App Store Review Cycles:** Apple App Store and Google Play Store review processes are governed by respective platform policies (typically 24 to 72 hours).
- **Payment Gateway Merchant KYC:** If optional UPI is chosen, activation of merchant live keys depends on Razorpay/Cashfree KYC verification of the client's business entity.

22 Out-of-Scope & 23 Future Roadmap

SCOPE BOUNDARIES

22. Out-of-Scope Items (Explicit Exclusions)

To eliminate ambiguity and prevent scope creep during execution, the following capabilities are explicitly identified as outside the scope of this project agreement:

Out-of-Scope Domain	Clarification & Explicit Boundary
Physical Hardware & IoT Locks	Physical procurement, cabling, or hardware-level installation of biometric fingerprint scanners, RFID readers, or IoT smart door locks.
On-Ground Police & Tenant Verification	Physical dispatch of tenant verification forms to local police stations or on-ground legal background screening. (The software provides digital document capture only).
ERP & Legacy Accounting Sync	Custom two-way automated synchronization with enterprise ERPs (e.g. SAP) or desktop accounting packages (Tally). Standard Excel export is provided.
Cafeteria & Kitchen Food Inventory	Raw food grocery inventory tracking, kitchen recipe cost calculation, or mess staff attendance tracking.
Third-Party Direct Expenses	Any ongoing subscription fees, cloud hosting charges, SMS gateway credits, or Meta conversation charges (payable directly by client to providers).

23. Future Enhancements Roadmap (Phase 2 & 3 Evolution)

1. IoT Smart Door Locks
Integration with Tuya/Zigbee smart locks, generating temporary digital passcodes for checked-in tenants that expire upon checkout.

2. Smart Sub-Meter Billing
Automated reading of smart electricity sub-meters per room, splitting utility bills automatically among sharing roommates on monthly rent invoices.

3. AI Dynamic Pricing
Machine learning algorithms analyzing seasonal university intake cycles and local corporate hiring to recommend optimal room pricing.

4. Maintenance Tickets
Tenant mobile portal for raising repair requests (plumbing, Wi-Fi, AC) with photo attachments, SLA tracking, and warden assignment.

5. Multi-Branch Franchise
Franchise portal allowing independent property owners to list on a centralized aggregator brand with automated revenue-sharing splits.

6. Biometric Curfew Check
Automated evening attendance check via tablet facial recognition, dispatching automated WhatsApp safety alerts to parents if curfew is missed.

24 Maintenance, SLA & 25 Project Sign-Off

FORMAL ACCEPTANCE

24. Maintenance, SLA & Support Framework

The project includes **30 calendar days of complimentary post-launch warranty support** covering bug rectification, critical security patches, and deployment stabilization. Following warranty expiry, the client may elect an ongoing Annual Maintenance Contract (AMC):

Service Tier	Included Support Services & SLA Guarantees	Monthly Investment
Standard AMC	Cloud server monitoring, OS security updates, weekly database backups, and critical bug rectifications (SLA: <12h response).	**₹18,000 / month**
Comprehensive AMC	Standard AMC + 15 hours/month of feature enhancements, monthly optimization, priority SLA (<2h response for critical issues).	**₹32,000 / month**

25. Conclusion and Formal Sign-Off

The **PG Booking & Management Platform** delivers a production-grade, highly scalable solution addressing the critical operational bottlenecks of modern co-living management. By coupling conversational WhatsApp engagement with granular bed-level inventory tracking and modular financial processing, the platform positions the client for rapid multi-property operational expansion.

By signing below, the authorized representatives approve this project specification report, scope boundary, milestone roadmap, and cost estimation:

CLIENT ORGANIZATION

Client Representative Sign-Off

Authorized Signatory: _____
Designation: _____
Date: _____

DEVELOPMENT PARTNER

Technology Lead Sign-Off

Technical Director: _____
Designation: Head of Solutions Architecture
Date: _____

🚀 **Ready for Project Initiation**

Upon signature execution and receipt of the Phase 1 commencement deposit, the development team will commence Phase 1 discovery and sprint setup within 3 business days.